

Isha Andrade

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ABOUT ME

Driven Computer Science Engineering graduate and AI Research Intern at Coding Jr., with expertise in machine learning, deep learning, and data science. Skilled in building and optimizing AI models to solve complex problems across multiple domains, with a proven ability to thrive in fast-paced startup environments. Passionate about innovation, research, and the intersection of AI and music, dedicated to developing technology that enhances music learning and creativity.

EDUCATION AND TRAINING

B.E. (Hons.) Computer Science

Birla Institute of Technology & Science, Dubai [2020 – 2024]

City: Dubai | Country: United Arab Emirates | Website: <https://www.bits-pilani.ac.in/dubai> | Final grade: 9.35 / 10

Relevant Coursework: Computer Programming, Object Oriented Programming, Data Structures & Algorithms, Database Systems, Logic in Computer Science, Theory of Computation, Mathematics (I, II, III), Probability & Statistics, Discrete Structures for Computer Science, Computer Architecture, Computer Networks, Design and Analysis of Algorithms, Compiler Construction, Operating Systems, Digital Design, Microprocessors & Interfacing, Machine Learning, Artificial Intelligence, Foundations of Data Science, Natural Language Processing

All India Senior School Certificate Examination (AISCE)

St. Joseph's School, Abu Dhabi [2018 – 2020]

City: Abu Dhabi | Country: United Arab Emirates | Website: <https://www.stjosephsschool.ae> | Final grade: 92.2%

Core Subjects: Computer Science, Mathematics, Physics, Chemistry, English

All India Secondary School Examination (AISSE)

St. Joseph's School, Abu Dhabi [2016 – 2018]

City: Abu Dhabi | Country: United Arab Emirates | Website: <https://www.stjosephsschool.ae> | Final grade: 91%

Core Subjects: Mathematics, Science (Physics, Chemistry, Biology), English, French, Social Science

SKILLS

C / C++ / Java / Python / R / SQL / Flutter & Dart / Kotlin / Spring / Spring Boot / HTML / CSS / JavaScript / IDE: Jupyter Notebook, Google Colab, RStudio / Git & GitHub / Power BI / Machine Learning / Deep Learning / Artificial Intelligence / Natural Language Processing / Computer Vision / VS Code / IntelliJ / MongoDB / Express.js / Node.js / React / React-Native

PUBLICATIONS

[2024]

Intelligent Intrusion Detection Using ML for Large-Scale IoT Networks

This research addresses network security in large-scale IoT environments using machine learning. Four ML classifiers—logistic regression (LR), perceptron, AdaBoost, and random forest (RF)—were tested on the CICIoT2023 dataset. Results show that the RF classifier outperforms others with 99.75% accuracy for binary classification and 99.52% for multi-class classification. The RF model outperforms state-of-the-art ML techniques in detecting and classifying various IoT network attacks.

Presented at the 2024 Advances in Science and Engineering Technology International Conferences (ASET) held in Dubai, UAE, and published in IEEE Xplore.

WORK EXPERIENCE

 CodingJr

Website: <https://codingjr.school>

AI Research Intern

[01/2025 – Current]

- Conduct in-depth research on the tech stacks, business models, and AI integration opportunities of assigned companies to identify areas for improvement.
- Design and develop a tailored LLM-powered CoPilot that seamlessly integrates with each company's tech stack, automating tasks, providing actionable insights, and enhancing overall workflow efficiency.
- Analyze existing CoPilot solutions and evaluate alternative AI models to drive innovation and streamline business processes.

 Freelance

Python Web Scraping & Data Analyst

[11/2024 – 12/2024]

- Developed a Python-based web scraper to automate the extraction of case data from a legal website, with functionality to detect and track new and updated entries
- Integrated Google Sheets API to automatically record real-time updates, enabling the client to access live, up-to-date case data.
- Ensured secure management of sensitive client information using environment variables, delivering a fully automated solution that reduced manual data entry and improved workflow efficiency

 Raag (Early-stage Startup)

Founder

[06/2024 – Current]

- Ideated and conceptualized raag, an AI-driven platform to gamify music practice for users and offer personalized practice schedules tailored to each user's skill level and goals.
- Created and refined wireframes using Figma to outline the platform's user interface and user path

 First Response Healthcare – Dubai, United Arab Emirates

City: Dubai | Country: United Arab Emirates | Website: <https://firstresponsehealthcare.com>

Software Engineer Intern

[01/2024 – 07/2024]

- Developed an AI-driven Blood Pressure prediction feature for the company app, using computer vision (OpenCV) and signal processing libraries (SciPy) to analyze light intensity reflected from a user's face at periodic timestamps. The intensity data was utilized to construct a PPG signal graph, which was then processed by a CNN model trained on the PulseDB dataset to predict the user's real-time systolic and diastolic blood pressure.
- Collaborated with a team of developers using Git and Github to create the company's app, employing Flutter with BLoC state management for a responsive UI, and Spring Boot with Kotlin and Firebase Cloud Messaging for backend development.

 Birla Institute of Technology & Science, Dubai – Dubai, United Arab Emirates

City: Dubai | Country: United Arab Emirates

Object Oriented Programming Lab Assistant, CS department

[09/2023 – 12/2023]

- Strengthened students' understanding of various Object-Oriented Programming concepts through the use of Java.
- Provided personalized guidance, resolved lab queries, and debugged errors, enhancing overall student learning outcomes.

 Borouge (Abu Dhabi National Oil Company (ADNOC) Group) – Abu Dhabi, United Arab Emirates

City: Abu Dhabi | Country: United Arab Emirates

Intern

[06/2022 – 08/2022]

- Fostered strong collaboration with Borouge's computer science team by facilitating regular meetings, enhancing communication and teamwork skills
- Maintained timely task submissions and crafted detailed weekly progress reports, demonstrating strong time management and attention to detail

PROJECTS

Deep Learning for IoT Network Security

- Designed and implemented two novel Deep Learning models: a two-layer Convolutional Neural Network (CNN) and a two-layer Long-Short Term Memory (LSTM) model.
- Trained the models on the CICIoT2023 dataset, a real-time benchmark dataset encompassing various large-scale attacks in IoT (Internet of Things) environments, including DDoS and Brute force attacks.
- Conducted three types of classification tasks:
 - Binary Classification: distinguishing between malicious and benign network traffic.
 - Grouped Classification: categorizing network attacks into 7 groups.
 - Multi-class Classification: identifying 33 different types of IoT network attacks.
- Achieved exceptional accuracies with the proposed models:
 - CNN model: 99.34% (binary), 99.02% (grouped), and 98.62% (multi-class).
 - LSTM model: 99.42% (binary), 99.13% (grouped), and 98.68% (multi-class).
- Performed a comparative analysis with four traditional Machine Learning models (Logistic Regression, AdaBoost, Perceptron, and Random Forest), where Random Forest outperformed the other models with accuracies of 99.75%, 99.63%, and 99.51%.

Link: <https://github.com/ishaandrade/IoT-Network-Security>

Chess Engine using Reinforcement Learning

- Developed a chess engine using reinforcement learning (RL) techniques.
- Implemented a reward-based system where the computer learns to make moves based on favorable outcomes (winning the game).
- Trained the engine over time, observing its gradual improvement in performance.
- Achieved notable success, winning a total of 13 games after training for 3 hours, with continued improvement observed with extended training durations.

Link: <https://colab.research.google.com/drive/1Qq6VcQubXmdescujstGKMi8v6XRr9gcr#scrollTo=8oWhiq4mffWk>

Tesla Stock Price Prediction

- Cleaned, preprocessed, and enriched Tesla stock price dataset by handling missing values, removing outliers, and engineering features such as different-period simple and exponential moving averages for an improved prediction of Tesla Close prices
- Performed a comparative analysis of four standard ML models, namely, Linear Regression (LR), Random Forest Regressor (RFR), Gradient Boosting Regressor (GBR), and Support Vector Machine (SVM) which were trained on the processed Tesla Stock Price dataset to predict the future Close price. Out of the four ML models, LR has the highest R^2 score however the GBR model predicted the most accurate future close price.

Link: <https://github.com/ishaandrade/stock-price-prediction>

Mental Health Risk Assessment of Text Using Zero-shot and Few-shot Learning

- Used the BART-Large-MNLI transformer model to perform zero-shot entailment on a dataset containing social media texts from users to determine whether a user can be classified as having mental-health related issues
- Performed few-shot ($k = 1, 2, \dots, 10$) classification using on the same dataset using the Sentence Transformer (paraphrase-MiniLM-L3-v2) model
- Achieved accuracy, recall, precision and f1-score between the range 70 to 80% for both the zero-shot and few-shot classifiers with the zero-shot classifier achieving a slightly higher accuracy of 79%
- Implemented a simple real-world application utilizing this model with SMTP protocol and python which sent an email to the guardian of a user under 18 if they posted a concerning message in the input dialog box

Link: <https://github.com/ishaandrade/application-of-zero-shot-and-few-shot-learning>

Online Product Store using MERN Stack

- MERN Product Store is a full-stack web application built using the MERN stack (MongoDB, Express.js, React, Node.js).
- It allows users to create, view, update, and delete products with real-time UI updates and toast notifications.
- The frontend is developed with React + Vite, styled using Chakra UI, and uses Zustand for global state management.
- The backend is powered by Express and connects to MongoDB using Mongoose.
- The app supports light/dark mode and provides a smooth, responsive user experience.

Links: <https://mern-product-store-wx68.onrender.com> | <https://github.com/ishaandrade/mern-product-store>

ADDITIONAL TRAINING AND CERTIFICATIONS

[28/04/2025]

Python for Data Science, AI & Development issued by IBM

Produce timely and accurate solutions to Bitcoin-related coding assignments

Link: <https://coursera.org/share/05db8c984c2b5434739713ed4e460b0b>

[16/02/2025 – Current]

Summer of Bitcoin 2025 Bootcamp - Developer Track

[11/04/2024]

What is Data Science? issued by IBM

Link: <https://www.coursera.org/account/accomplishments/records/LAQSBPNR923E>

[27/02/2024]

Getting Started with Flutter Development issued by Google Cloud

Link: <https://www.coursera.org/account/accomplishments/verify/9ARLYHRJN5K3>

[10/12/2023]

Power BI in Data Analytics Certificate issued by PwC Switzerland

Link: [https://forage-uploads-prod.s3.amazonaws.com/completion-certificates/PwC Switzerland/a87GpgE6tiku7q3gu_PwC Switzerland_7tGtHz5wQ4e6HzpAx_1702190900781_completion_certificate.pdf](https://forage-uploads-prod.s3.amazonaws.com/completion-certificates/PwC%20Switzerland/a87GpgE6tiku7q3gu_PwC%20Switzerland_7tGtHz5wQ4e6HzpAx_1702190900781_completion_certificate.pdf)

[25/11/2023]

Neural Networks and Deep Learning issued by DeepLearning.AI

Link: <https://www.coursera.org/account/accomplishments/certificate/JBFG2JR4DXRX>

HONOURS AND AWARDS

[09/2023] Birla Institute of Technology & Science, Dubai

Summer Insurance Broking Game Development Competition

Result: 3rd Place

[03/04/2023] Trinity College London

Piano Grade 8

Result: Distinction

Link: <https://drive.google.com/file/d/1jfUt9nhQqhlvMfdv6CVDh4r-e5kWXNuo/view>